



Dysregulation of energy homeostasis in amyotrophic lateral sclerosis

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Purpose of review

Amyotrophic lateral sclerosis (ALS) is a neurodegenerative disease targeting upper and lower motor neurons, inexorably leading to an early death. Defects in energy metabolism have been associated with ALS, including weight loss, increased energy expenditure, decreased body fat mass and increased use of lipid nutrients at the expense of carbohydrates. We review here recent findings on impaired energy metabolism in ALS, and its clinical importance.

Recent findings

Hypothalamic atrophy, as well as alterations in hypothalamic peptides controlling energy metabolism, have been associated with metabolic derangements. Recent studies showed that mutations causing familial ALS impact various metabolic pathways, in particular mitochondrial function, and lipid and carbohydrate metabolism, which could underlie these metabolic defects in patients. Importantly, slowing weight loss, through high caloric diets, is a promising therapeutic strategy, and early clinical trials indicated that it might improve survival in at least a subset of patients. More research is needed to improve these therapeutic strategies, define pharmacological options, and refine the population of ALS patients that would benefit from these approaches.

Summary

Dysfunctional energy homeostasis is a major feature of ALS clinical picture and emerges as a potential therapeutic target.

Keywords

amyotrophic lateral sclerosis, diet, energy metabolism, hypothalamus, weight loss

INTRODUCTION

Until recently, amyotrophic lateral sclerosis (ALS) was considered a disease restricted to the motor system. The simultaneous degeneration of upper motor neurons, in the motor cortex, and of lower motor neurons, in the brainstem and spinal cord, appeared sufficient to explain the clinical phenotype of patients. Recent years have however demonstrated that ALS signs and symptoms are not restricted to the motor system, but also involve cognitive and metabolic alterations. This idea is consistent with the notion that ALS is part of a continuum with frontotemporal dementia (FTD) [1,2].

More than two decades ago, Couratier *et al.* observed that malnutrition correlated with worsened survival of ALS patients [3]. These observations, underpinned by studies in animal models [4], highlighted the alteration of systemic energy homeostasis in ALS patients. In recent years, an important body of evidence documented the clinical importance of the dysfunctional energy homeostasis observed in ALS. Anatomical and cellular substrates

for these mechanisms are still poorly understood. Here, we review details of these emerging concepts in ALS.

EVIDENCE OF SYSTEMIC ENERGY HOMEOSTASIS ABNORMALITIES IN AMYOTROPHIC LATERAL SCLEROSIS

The occurrence of premorbid weight loss in ALS is the most evident symptom of defective energy

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Motor neuron disease

KEY POINTS

- Weight loss is a major symptom of ALS, occurring early, independently of dysphagia and not fully explained by denervation-induced muscle atrophy.
- Causes of weight loss are incompletely characterized and include alterations in metabolic fluxes in various tissues including skeletal muscle.
- Hypothalamic atrophy, as well as alterations in hypothalamic peptides controlling energy metabolism, have been associated with metabolic derangements in ALS.
- High caloric diet is a promising therapeutic strategy, with preliminary evidence of efficacy in fast progressing ALS patients.

metabolism. ALS patients usually display normal to low body mass index at onset [5], and typically lose weight and body fat with the progression of the disease [3]. However, weight loss is not observed in all ALS patients, affecting between one and two-thirds of individuals [6^a,7,8]. Consistent with the importance of weight loss in the disease process, the risk of developing ALS has been repeatedly shown to increase with lower premorbid body fat [9–11] and is also correlated with lower levels of leptin, an adipocyte-derived hormone reflecting adipose energy stores [12]. The process of weight loss in ALS appears to precede the onset of motor symptoms as presymptomatic ALS patients begin losing weight several years before the disease onset and diagnosis [13]. The systemic metabolic impairment in ALS remains incompletely characterized, and several studies have documented paradoxical glucose intolerance [14], insulin resistance [15] and, at least in some studies, increased circulating lipids [16,17] as well as redistribution of adipose tissue towards more visceral fat [18].

Weight loss appears to affect patients with spinal and bulbar onset of symptoms [6^a], and, importantly (see later) occurs also in a significant proportion of patients who do not have the problem of dysphagia [6^a,19]. Weight loss is also not fully explained by denervation-induced muscle atrophy. This reinforces weight loss is not a mere consequence of disease progression but rather an initial symptom. Furthermore, the weight alteration (as low as 5% of initial body weight) is adversely associated with the survival of ALS patients either in the total ALS population [6^a,20–22] or after invasive ventilation [23]. This deleterious effect of weight change is observed in various genetic backgrounds,

either European or Asian populations [19,24], which makes it an important prognostic factor [6^a].

MECHANISMS OF SYSTEMIC ENERGY METABOLISM IMPAIRMENT

What causes weight loss in ALS patients? Weight loss could be caused either by increased energy expenditure or decreased energy intake [4] (Fig. 1). Interestingly, premorbid physical activity, that increases energy expenditure, has been repeatedly associated with ALS risk [25,26].

Besides, a large proportion of ALS patients display higher than predicted resting energy expenditure before any intervention with ventilatory support [27,28]. In 2021, it is accepted that hypermetabolism is observed in ALS patients [29] and mouse models [30,31], and is correlated with greater functional decline [32,33]. On a theoretical note, however, it is important to realize that the concept of hypermetabolism has been discussed for many years and has to be considered with caution, particularly in animal models. The relationship between energy expenditure and body weight has to be used with particular caution [34]. The traditional normalisation by body weight in small mammals is questionable, whereas the Harris and Benedict equation, used in humans, considers several physiological parameters and describes more accurately energy expenditure [35,36]. A re-evaluation of hypermetabolism in ALS is therefore necessary.

Decreased energy intake is a mechanism observed during the progression of ALS symptoms, which ultimately causes difficulties in eating and changes in dietary habits, resulting from dysphagia (Fig. 1). However, more recent studies converge to show that weight alteration occurs in patients even in the absence of early dysphagia [6^a,19]. Studies quantitating the appetite of ALS patients using appetite questionnaires demonstrated that ALS patients display loss of appetite [37], which is unrelated to bulbar symptoms and dysphagia [38]. Most importantly, this loss of appetite worsens with disease progression and is correlated with loss of weight and fat mass [38,39]. These changes in eating behaviour are related to cognitive defects and distinguish between patients with pure ALS and those with cognitive impairment [40]. It cannot be excluded that neuromuscular respiratory failure could also in itself, possibly through CO₂ retention or hypoxia, cause loss of appetite [41].

Weight and appetite are controlled centrally and these pathways are affected in ALS [42,43^a]. The key brain region in the process is the hypothalamus which finely controls the energy homeostasis [44] (Fig. 1). Arcuate nucleus neurons such as pro-

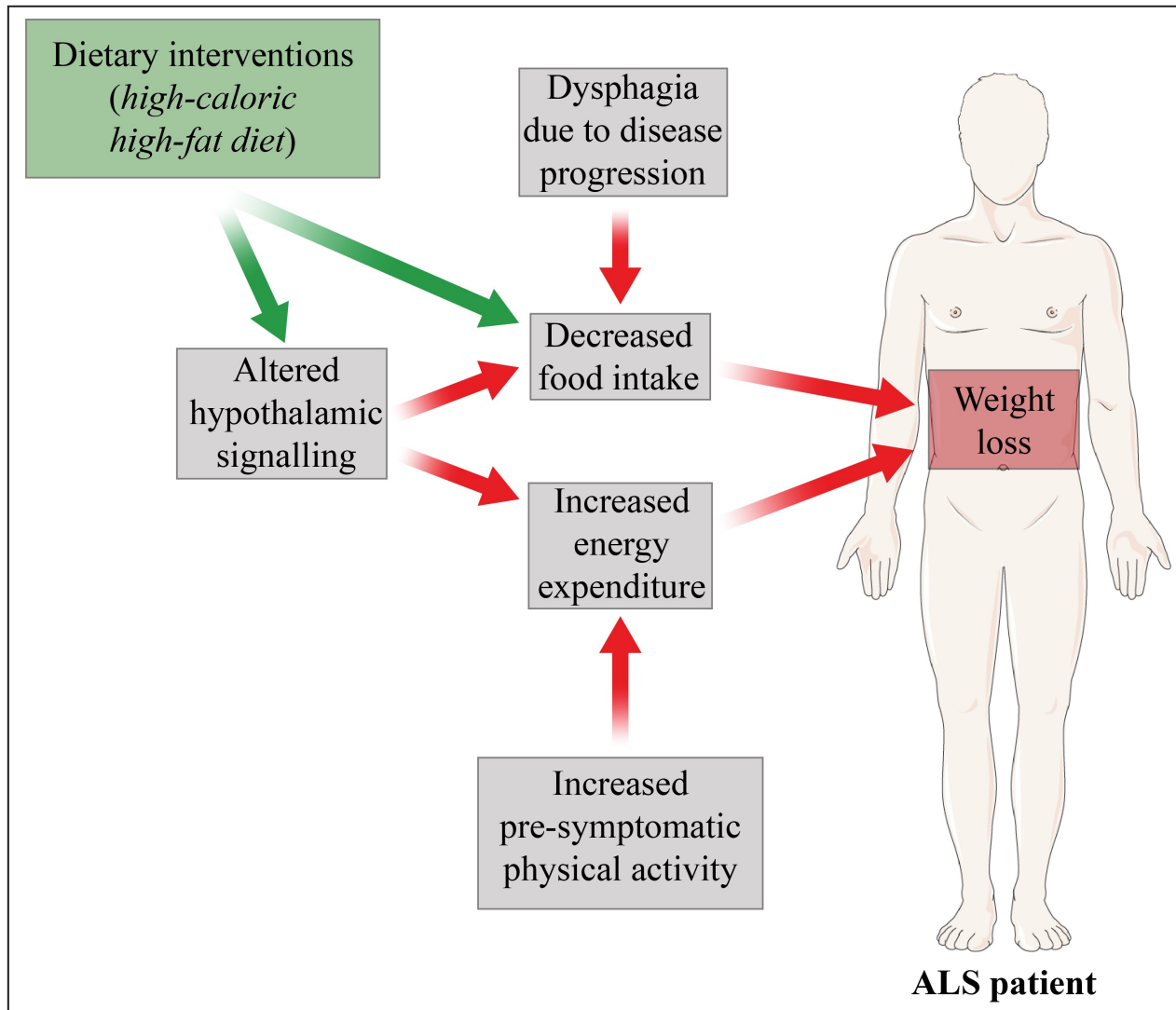


FIGURE 1. Mechanisms of weight loss in ALS patients. ALS, amyotrophic lateral sclerosis.

opiomelanocortin (POMC)- and agouti-related peptide (AgRP)-expressing cells are the first-order neurons responsive to energy status, and respectively decrease or increase food intake. Interestingly several studies showed that the hypothalamus accumulates TDP-43 lesions [45,46²²], especially the lateral hypothalamic area [46²²] (Fig. 2). Indeed, the hypothalamus appears atrophied, as measured using Magnetic resonance imaging [47], and hypothalamic connectivity is altered in both patients and ALS mice [48²²]. Functional evidence linking hypothalamic neuro-modulators to weight loss remains scarce. Indirect evidence pointed out that the hypothalamus of ALS patients does not adequately respond to food intake inducing cues. ALS patients exposed to the antidiabetic drug, pioglitazone, did not gain weight, whereas this drug acts selectively on hypothalamic POMC neurons to increase body weight. Consistently,

pioglitazone did not increase food intake in ALS mice [49]. Such blunted pioglitazone response is likely due to a defect in melanocortineric neurons, as POMC expression is decreased and AgRP expression is increased in ALS mice [49]. Independent studies showed increased levels of neuropeptide Y, a neuropeptide co-expressed with AgRP, in ALS patients [15].

The hypothalamic involvement in ALS remains, however, to be disentangled, and a recent study showed a decreased orexin expression in lateral hypothalamus neurons, that could contribute to weight loss [46²²] (Fig. 2). Further functional investigation of these hypothalamic neuronal subtypes in ALS weight loss is warranted [50]. However, ablation of leptin, an adipocyte-derived hormone acting on the hypothalamus to induce satiety, in mutant mice expressing the *SOD1* ALS-related mutation, increased lifespan and improved muscle function

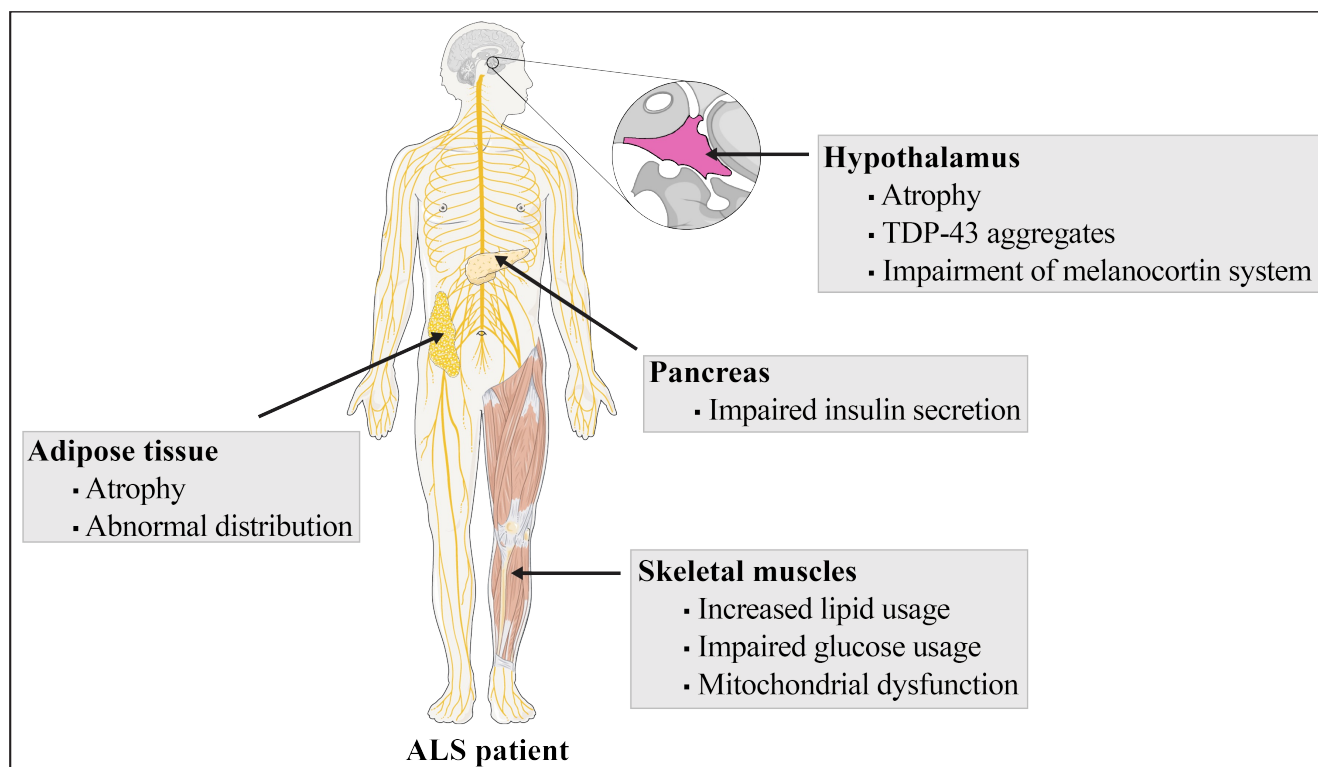


FIGURE 2. Systemic metabolic alterations in ALS patients and models. ALS, amyotrophic lateral sclerosis.

[31], indirectly suggesting the relevance of the hypothalamus to ALS progression.

ABNORMAL METABOLIC FLUXES IN AMYOTROPHIC LATERAL SCLEROSIS

Abnormal usage of nutrients might underlie weight loss. In the skeletal muscle of a mouse model expressing mutant SOD1, Palamiuc *et al.* showed that lipid metabolism was favoured over glucose at an early stage of the disease onset (Fig. 2). Consistently, mutant SOD1 mice showed an increased mitochondrial β -oxidation, indicative of increased lipid catabolism [51[■]]. Early in the disease progression a decreased respiratory exchange ratio [30,51[■]], altered gene expression [30,51[■],52] and an increased oxidative capacity [51[■]] were described. The increased lipid catabolism is conversely associated with decreased glucose oxidation. Thus, mutant SOD1 mice develop progressive glucose intolerance as a consequence of reduced usage of muscle glucose [51[■],52,53]. Interestingly, *SOD1* is not the only ALS related gene whose manipulation in transgenic animals modifies the usage of metabolic fuels. The overexpression of mutant A315T TDP-43 leads to increased adipose tissue and abnormal muscle response to insulin [54], whereas conditional knock-out of TDP-43 dramatically increased the usage of lipids. The latter results in loss of adipose tissue and death within days

[55]. Such alterations in fuel usage may be relevant to ALS patients since a similar increase in β -oxidation was observed in patients-derived primary myotubes [56[■]] and in patient-derived fibroblasts [57]. Further-more, these altered metabolic fluxes would be con-sistent with the trend of ALS patients to develop paradoxical insulin resistance as well as with the protective effects of increased blood lipids in ALS [17,58]. Moreover, cells carrying the *C9ORF72* mutation also exhibited substrate-specific alterations of bioenergetic substrates, in particular for pyruvate and glycogen [59]. These altered metabolic fluxes could constitute a compensatory response to abnormal mitochondrial function [30,51[■]]. Multiple pieces of evidence point to mitochondrial damage in response to mutant proteins associated with ALS, such as SOD1, TDP-43 and Fused in sarcoma (FUS) [60–62], particularly in muscles and axons. These impaired mitochondria in muscles are located at the neuromuscular junctions [63–65] which suggest that they participate in neuromuscular junction demise [61,66,67].

HOW COULD ALTERED METABOLIC FLUXES PARTICIPATE IN AMYOTROPHIC LATERAL SCLEROSIS?

Pharmacological evidence suggests that the metabolic shift towards lipid use participates in disease

progression. Dichloroacetate, an inhibitor of pyruvate dehydrogenase kinase, improves mitochondrial functions in mutant *SOD1* mice resulting in a slower weight loss [52]. Furthermore, ranolazine, a compound known to decrease β -oxidation was able to temporarily restore metabolic homeostasis, although it did not improve survival [51[■]]. Strikingly, downregulating the key enzyme responsible for fatty acid import into mitochondria, carnithine palmitoyltransferase 1, was able to slow down disease progression [68[■]].

Genetic interventions aiming at genetically boosting mitochondrial biogenesis, through overexpression of peroxisome proliferator-activated receptor- γ coactivator 1- α (PGC-1 α), was able to prevent muscle atrophy and improve mitochondrial function by restoration of muscle signalling and ATP consumption [69]. Conversely, knocking out a major muscle isoform of PGC-1 α was detrimental to male mutant *SOD1* mice [70]. The effects of PGC-1 α overexpression appear however discordant across studies, and, more recent evidence suggests that overexpressing PGC-1 α would not prevent or delay the onset of the disease [30,71].

An alternative strategy to manipulate metabolic fluxes could be to favour glucose usage. A recent study demonstrated that increased expression of GLUT-3 (glucose transporter 3) and phosphofructokinase in TDP-43 fly models was neuroprotective in the nervous system, but not in muscles [72]. Similar results were obtained in fly and cell models of *C9ORF72* ALS through the improvement of insulin signalling [73]. This indirectly suggests that improvement of glucose metabolism could modify disease progression. Such therapeutic strategies aiming at reorientating energy metabolism to glucose usage have however not yet been translated through the clinic.

DIETARY MODULATION OF METABOLISM IS EFFICIENT IN MODELS AND PATIENTS

Dietary improvement of patients could also potentially modify energy metabolism (Fig. 1). In 2004, we reported that a high-fat diet could improve metabolic status, prevent the loss of motor neurons, and extend survival in mutant *SOD1* mice [30]. More recently, similar results were established in TDP-43 mutant mice using a high-fat diet to show reduced oxidative stress and improved presymptomatic state [74]. It was shown recently however that very high-fat content was associated with poorer outcomes in mutant *SOD1* mice [68[■]], suggesting a bell-shaped curve in the protection mediated by high caloric intake in mouse models.

The first clinical trial was performed by Wills and collaborators, which observed improved survival in

gastrostomised patients fed with high caloric diets as compared with an isocaloric diet [75,76]. Lately, the lipid supplementation in LIPCAL-ALS trial observed a protective effect of dietary lipid supplementation on weight loss and survival of fast progressing ALS patients [77[■]], although the trial did not meet its primary endpoint on the total ALS population. This dietary intervention decreased circulating neurofilament levels, consistent with a neuroprotective action [78]. Further studies, in particular with prespecification of the analysis of fast-progressing patients, are warranted to confirm these initial studies. Other dietary interventions could prove useful in ALS but have not reached the stage of clinical studies. Ketone bodies or ketogenic diets have shown potential in mutant *SOD1* mice [79] and fly models of TDP-43 ALS [80], and remain to be further tested.

Several studies have suggested that intake of polyunsaturated fatty acids could protect against ALS [81,82]. However, no clinical trial has yet tested whether there could be differences in efficacy depending on the degree of unsaturation.

HOW ARE AMYOTROPHIC LATERAL SCLEROSIS-RELATED MUTATIONS INTERFERING WITH ENERGY METABOLISM?

ALS is a disease with a prominent genetic contribution, with 4 major genes (*SOD1*, *C9ORF72*, *TARDBP* encoding TDP-43 and *FUS*). How these genetic mutations cause or contribute to metabolic defects remain incompletely characterized. Mutations in *SOD1* were the first genetic causes of ALS identified, and most experimental studies still use transgenic mice expressing mutant forms of *SOD1*. How mutant *SOD1*s might precipitate weight loss and shift energy metabolism remains unclear. Several lines of evidence suggest that *SOD1* mutant proteins might affect mitochondrial function, through interaction with major mitochondrial proteins such as VDAC1 [83] or Bcl-2 [84], leading to oxidative stress, damage to mitochondrial RNA [85] and mitochondrial damage [86–90]. Further work should dissect the relative contributions of these different mechanisms, and their respective contributions in specific cell types.

As for *SOD1*, mutations in *TARDBP*, encoding TDP-43 appear to have important effects on mitochondrial physiology, that could possibly underlie dysfunction in energy homeostasis. Indeed, aggregation of TDP-43 has been reported to sequester several key mRNAs which encode nuclear-encoded mitochondrial proteins [91]. A fraction of TDP-43 enters the mitochondria and affects mitochondrial function through different mechanisms including

alterations in mitochondrial transcript metabolism, activation of the mitochondrial transition pore or the mitochondrial unfolded protein response [92–94]. TDP-43 might also have effects on energy metabolism beyond mitochondria and has been shown to directly regulate insulin secretion (Fig. 2) through regulation of a circular RNA encoded by the insulin gene [95[■]] and through regulation of the expression of the calcium channel CaV1.2 in pancreatic β -cells [96[■]].

Like TDP-43, FUS is an RNA binding protein involved in multiple aspects of RNA metabolism in both ALS and ALS-FTD. FUS has been reported to regulate the stress response and mitochondrial functions, such as transport and autophagy processes [97]. FUS modulates mitochondrial oxidative stress in muscles by inhibiting the action of the heat shock protein 60 [97,98]. The *FUS* gene appears to encode for a second polypeptide, called altFUS, that is mitochondrially targeted and required for the toxic effects in *FUS* mutants [99].

Expansion of a hexanucleotide repeat in the *C9ORF72* gene is associated with both ALS and ALS-FTD and leads to motor neuron toxicity through multiple mechanisms, including haploinsufficiency of the *C9ORF72* gene product and toxicity of RAN-translated di-peptide protein repeats. Although defective mitochondria have been found in both mutant models and ALS and ALS-FTD patients carrying the *C9ORF72* mutation [100,101], knowledge about those mitochondrial alterations are sparse [102]. It has been shown that the *C9ORF72* protein might be involved in stabilizing mitochondrial complex I [103]. It remains however unclear how the hexanucleotide repeat expansion leads to mitochondrial defects.

Beyond familial ALS, a large proportion of ALS cases do not have familial history. *ACSL5* was recently highlighted in a genome-wide association study as a genetic factor increasing ALS risk. *ACSL5* is known to convert free long-chain fatty acids into fatty-acid-coenzyme A, thereby, playing a key role in lipid biosynthesis and fatty acids degradation [61]. Overexpression of *ACSL5* was associated with rapid weight loss, as has *GPX3*, another ALS-related gene [104,105,106[■]]. Additional loci, including *B4GALNT1*, *G2E3-SCFD1* and *TRIP11-ATXN3*, have been identified with a gene-based analysis as disease-associated genes [106[■]]. This study suggests that there are genetic contributions to the observed alterations in energy metabolism in ALS patients, beyond familial ALS. Further investigations are required to identify additional mechanisms and decipher their role and contributions to the observed metabolic defects.

CONCLUSION

Weight loss has emerged in the last ten years as a strong prognostic marker in ALS. Clinical, epidemiological, and experimental evidence all converge to highlight the importance of this process in disease progression. The recent clinical studies are very encouraging in pursuing nutritional therapeutic trials and identifying anatomical and biological elements involved in weight loss, as well as subpopulations of patients that might benefit from dietary adaptations. Dietary correction of the dysregulated energy metabolism in ALS is likely to emerge as a future disease-modifying approach in ALS.

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Conflicts of interest

There are no conflicts of interest.

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